Project Report: Temple Visitor Analytics System

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1. Executive Summary

We have successfully developed and optimized a custom Artificial Intelligence surveillance pipeline to monitor, count, and classify temple visitors. The traditional approach of using "tripwires" (gate-lines) fails in temple environments due to high occlusion, workers sitting in place, and loitering.

To solve this, we architected a hardware-accelerated **Skip-Frame Re-Identification (Re-ID) Pipeline**. This system mathematically fingerprints every individual entering the frame and tracks them dynamically, completely eliminating double-counting and the need for gate-crossings.

2. Artificial Intelligence Architecture

Our pipeline relies on three cascaded deep-learning models, optimized specifically for NVIDIA Tensor Cores:

1. **Detection (YOLOv8s):** A PyTorch-based spatial detector bounding all humans in the frame. It ignores rigid background motion and is paired with a **Zone Filter** to explicitly blind the AI to designated "worker-only" or "restricted" zones.
2. **Identity Tracking (OSNet-x1_0):** A Convolutional Neural Network that extracts a 512-dimensional vector embedding for every person. Using Hungarian matching and Cosine Distance matrices, it links individuals frame-to-frame.
3. **Demographic Classification (ConvNeXt-Tiny):** Compiled to ONNX (Open Neural Network Exchange), this model crops the detected human and predicts gender. To prevent flickering, it strictly requires 5 positive chronological votes before permanently "locking" a visitor's demographic assignment.

3. Engineering & Performance Optimizations

Processing high-definition CCTV footage normally requires massive server clusters. By engineering the following constraints into the pipeline, we achieved local hardware processing that outperforms real-time video speeds:

- **Hardware Acceleration:** Full utilization of CUDA Cores via PyTorch and `onnxruntime-gpu`

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- **Zero-Copy Skip Frames:** Analyzing video at 25 FPS is redundant for slow-moving crowds. The pipeline mathematically processes every 5th frame (0.2 seconds). Skipped frames bypass CPU memory allocation entirely.
- **Exponential Moving Average (EMA) Banks:** Instead of storing thousands of images of a person to remember them, the system mathematical blends their dynamic features into a single evolving 512-dimension vector.
- **Asynchronous Threaded Queue:** Writing annotated video to a hard drive bottlenecks ML processing. Output frames are pushed to a background thread to be compiled asynchronously, ensuring the GPU never stalls.

4. Performance Benchmarks

Tested locally on an NVIDIA RTX 3050 Ti Laptop GPU:

Metric	Result
Video Resolution	1080p HD
Pipeline Inference Speed	**~85 FPS**
Real-Time Multiplier	**3.5x Faster than Real-Time**
1- Hour CCTV Video	**17 Minutes to Process**
24-Hour CCTV Block	**~6.8 Hours to Process**

5. Output Deliverables

The client receives a complete analytical suite:

1. **Cumulative Global Count:** Total unique daily visitors, actively omitting returning staff.
2. **Demographic Breakdown:** Male vs. Female ratios.
3. **High-Traffic Timestamping:** The system monitors velocity spikes in the unique visitor count, automatically flagging specific timestamps as "High Density" events.
4. **Interactive Dashboard:** A local React/Next.js dashboard dynamically visualizes the Server-Sent Events (SSE) data in real-time.